Alagille Syndrome, Part Five: The Vessels. Intracranial Aneurysms, Moyamoya, Haemorrhage, and Whom to Screen with Magnetic Resonance Angiography, and When.

Imagine a girl of six with Alagille syndrome, confirmed by a pathogenic variant in the gene JAG1. She still has her own liver. Her total bilirubin is about three milligrams per decilitre while she takes an ileal bile acid transporter inhibitor, the class of drug that blocks the reabsorption of bile acids in the last part of the small intestine and is now widely used for the itch of this disease. She has narrowing of her branch pulmonary arteries, the problem discussed in the previous lesson. At today's visit her blood pressure is one hundred and twenty-eight over eighty-four millimetres of mercury, which is above the ninety-fifth centile for a child of her age and height. Her international normalised ratio, the standard measure of how fast the blood clots, is one point four. An hour ago she fell from a climbing frame and hit her head, and since then she has vomited twice. She has never had any imaging of her brain.

Her team has five questions. Does she need a computed tomography scan of the head tonight, and what does a slightly prolonged clotting time change? Should every child with Alagille syndrome have a magnetic resonance angiogram, usually shortened to MRA, which is an MRI scan tuned to show the arteries, and if so at what age? If such a scan were to show a narrowed carotid artery, or a small aneurysm on the basilar artery, who should act, and how fast? Is her high blood pressure coming from her kidneys, her aorta or her renal arteries? And how does all of this change her future: climbing and football, a liver transplant one day, a pregnancy one day?

The main idea of this chapter is that the genetic defect in Alagille syndrome produces a disease of the arteries throughout the body. Outside the heart, it narrows, stretches and balloons the arteries of the brain, the kidneys and the aorta. In the largest international cohort, non-cardiac vascular events caused about one in seven deaths, at a median age of only two years. A bleed usually needs two things at once: a weak vessel, and a push, such as trauma, vitamin K deficiency, high blood pressure or a procedure. No professional society tells us whom to scan. The best published expert practice is to scan the brain at about the age of eight, before major surgery or transplant, at the transition to adult care, and immediately for any neurological symptom.

One weak arterial wall, five ways to fail

The protein JAG1 is a ligand, a signalling molecule on the surface of one cell that activates Notch receptors on the cell next to it. In most patients with Alagille syndrome

one copy of JAG1 is lost, so the signal runs at roughly half strength. In the developing artery, Notch signalling tells the surrounding mural cells to become mature smooth muscle, the layer that gives an artery its strength. Mouse work published in 2007 in the Journal of Clinical Investigation by Frances High and colleagues showed that blocking this signal in the cells that build the great arteries produces narrowed pulmonary arteries and outflow defects. The same principle applies to the arteries of the brain and the abdomen: half the signal leaves a wall that is thinner and less well organised than it should be.

It helps to think of five ways this weak wall can fail. First, where the wall thickens or fails to grow, the artery narrows. This happens in the internal carotid arteries, the basilar artery, the renal arteries and the aorta. If narrowing of the end of the internal carotid artery progresses slowly, the brain grows a fine network of small collateral vessels around it, and the angiogram takes on the pattern called moyamoya, a Japanese word for a puff of smoke. Second, where the wall yields rather than thickens, the artery stretches. A long, widened and tortuous artery is called dolichoectasia, and in one radiology series three of four children with dolichoectasia also had saccular aneurysms, which are berry-like pouches on the artery wall. Third, there are the pushes that act on this fragile wall. Cholestasis, the reduced flow of bile, impairs the absorption of fat-soluble vitamin K, and vitamin K is needed to make several clotting factors. Portal hypertension, raised pressure in the vein that carries blood to the liver, enlarges the spleen, which traps and lowers platelets. Narrowing of a renal artery makes the kidney release renin and raises the blood pressure. And falls add mechanical shear. Fourth, a bleed follows when a fragile vessel meets one of these pushes. And fifth, some bleeds occur with no lesion found at all.

The practical meaning is that the vessel in the brain is the same vessel as the branch pulmonary artery we discussed in the last lesson. That explains why the lesions are often multiple, and why narrowed and dilated segments can coexist in one patient. It also gives prevention five levers, and only one of them is imaging. The others are vitamin K, platelets and the planning of procedures, blood pressure, and the prevention of head injury.

How common vascular disease is outside the heart

The foundational description is a 2004 paper in Circulation by Binita Kamath and colleagues from the Children's Hospital of Philadelphia. They reviewed two hundred and sixty-eight people with Alagille syndrome, followed between 1992 and 2002, and found non-cardiac vascular anomalies or events in twenty-five of them, about nine percent, or roughly one in eleven. Nine of these twenty-five had an intracranial event, such as a bleed or a stroke, without any structural lesion being found. Seven had an anomaly of the internal carotid artery: four had stenosis, two had agenesis, meaning the artery had never formed, and one had dolichoectasia. Three had aneurysms of the aorta, two had

aneurysms of the basilar artery, two had coarctation, a narrowing of the aorta, one had an aneurysm of the middle cerebral artery, one had moyamoya, and one had renal artery stenosis. The ages at which these problems appeared ranged from four months to forty-four years. Seventeen of the twenty-five carried a JAG1 variant themselves or had a first-degree relative who did.

Two details from that study are worth remembering. Among the nine people who had an intracranial event without a visible lesion, two had a head injury beforehand and five were critically ill or had other problems such as hypertension. And the mortality figures were striking. Of twenty-nine deaths in the cohort, ten, or thirty-four percent, were caused by non-cardiac vascular events. That was more than the twenty-one percent caused by heart disease and far more than the ten percent caused by liver disease. An earlier Philadelphia cohort of ninety-two patients, published by Karan Emerick and colleagues in Hepatology in 1999, had already found that intracranial bleeding caused about a quarter of deaths. The current GeneReviews chapter on Alagille syndrome, updated in 2024, cites neurovascular accidents in up to fifteen percent of patients.

The practical lesson is that a head bleed in a child with Alagille syndrome should be treated as a vascular event until proven otherwise, even when the first scan shows no aneurysm.

When vascular deaths happen

The best modern picture comes from the Global ALagille Alliance, known as GALA, an international registry that published the natural history of one thousand four hundred and thirty-three children in Hepatology in 2023, with Shannon Vandriel as first author. There were one hundred and eight deaths. Liver disease or complications of liver transplantation accounted for twenty-two percent, at a median age of two point eight years. Cardiac complications accounted for eighteen percent, at a median age of just over one year. Multiorgan failure accounted for fifteen percent. And non-cardiac vascular complications also accounted for fifteen percent, at a median age of two point two years, with half of these deaths falling between about one and a half and three years.

Put simply, in the largest cohort we have, vascular deaths are as frequent as deaths from multiorgan failure and nearly as frequent as cardiac deaths, and they happen years before the age at which most centres first think about screening. This is the central tension of the topic, and we will return to it when we discuss the timing of screening.

Bleeding without liver failure

A second line of evidence, from Paris, shows that children with Alagille syndrome bleed more than their liver disease would explain. In 2003, Panayotis Lykavieris and colleagues reported in Pediatrics on one hundred and seventy-four children with Alagille syndrome.

Thirty-eight of them, or twenty-two percent, had a significant bleed despite not having liver failure. A significant bleed was defined as a fall in haemoglobin of at least three grams per decilitre or a need for transfusion. Together these children had forty-nine bleeding episodes, at a median age of about three and three quarter years. Seventeen children bled spontaneously, and twenty-one bled during surgery or other procedures. Nine bled at least twice, and twenty-three needed a transfusion.

Two findings make this study important. First, the median platelet count and the median prothrombin time were normal, so standard tests did not predict who would bleed. Most of the children, thirty-three of the thirty-eight, had severe cholestasis. Second, eight children died from bleeding: four after surgery, two from bleeding in the gut, one after a needle liver biopsy, and one from an intracranial bleed.

The practical point is that a normal INR and a normal platelet count do not clear a child with Alagille syndrome for a biopsy or an operation. Before any procedure, ask whether it is truly needed. Give vitamin K, cross-match blood, choose the most experienced operator, and observe the child for longer than usual afterwards. The same fragility is why a child with Alagille syndrome who hits their head deserves a low threshold for imaging.

Five levers for prevention

It is worth turning the mechanism into a checklist, because most of what prevents a bleed happens in the hepatology clinic, not in the scanner. The first lever is vitamin K. Cholestatic malabsorption lowers the vitamin K dependent clotting factors, and the INR can drift between visits. Check it at each visit, supplement vitamin K, and give it by injection or infusion before procedures and after a head injury. The second lever is platelets. Portal hypertension and an enlarged spleen lower the platelet count, so count platelets before procedures, protect the spleen during sport, and avoid non-steroidal anti-inflammatory drugs. The third lever is blood pressure. Renal artery stenosis, mid-aortic syndrome, which is a long narrowing of the abdominal aorta, coarctation and renal dysplasia can all raise it. Measure it at every visit, in both arms, and image the renal arteries if it is high. The fourth lever is trauma. Remember that two of the nine lesion-free intracranial events in the Philadelphia series followed trauma. GeneReviews advises that people with Alagille syndrome avoid contact sports, and a child who vomits or becomes drowsy after a head injury should have a scan. The fifth lever is the lesion itself. Aneurysms, moyamoya and stenoses are often silent until they bleed or cause a stroke, and finding them is the job of screening. A good summary rule is to correct the coagulation, measure the pressure, prevent the fall, and only then argue about the scan.

What screening finds

Several series have used magnetic resonance angiography to look for intracranial lesions. The first systematic one was published by Karan Emerick and colleagues in the Journal of Pediatric Gastroenterology and Nutrition in 2005. Twenty-six patients were studied: twenty-two without symptoms and four with neurological symptoms. Cerebrovascular abnormalities were found in ten of the twenty-six, or thirty-eight percent. All four symptomatic patients had an abnormality, compared with five of the twenty-two asymptomatic patients, which is twenty-three percent, or nearly one in four. Most lesions involved the internal carotid arteries, and there was one basilar and one middle cerebral artery aneurysm.

A 2018 series from Cincinnati by Candice Carpenter and colleagues, published in Pediatric Radiology, reviewed fifty-two patients, of whom nineteen had dedicated vascular imaging. Six of those nineteen, about a third, had arterial disease, including four with dolichoectasia, three with aneurysms and two with moyamoya. Four of nineteen had venous anomalies, among them a persistent falcine sinus, a fetal venous channel that normally closes, which the authors estimated occurs in about six percent of people with Alagille syndrome compared with one to two percent of the general population. The authors concluded that non-invasive screening of the brain vessels should be performed in this population.

The most recent series, published in Neuroradiology in 2024 by Cerron-Vela and colleagues from Philadelphia, included thirty-two children with a median age of six years. Eighteen of the thirty-two, fifty-six percent, had some cerebrovascular abnormality, and the commonest category was stenosis or narrowing, in twelve, or thirty-eight percent. A review in the British Journal of Radiology by D'Amico and colleagues estimates the overall prevalence of arterial and venous anomalies at thirty to forty percent.

Two caveats matter. Every one of these series studied a selected group of children who happened to be imaged, so the true prevalence in an unselected population is probably lower. And for comparison, a large meta-analysis by Vlak and colleagues in Lancet Neurology in 2011 found unruptured intracranial aneurysms in about three percent of adults without specific risk factors, mostly in people over fifty. So the right way to read these numbers is that most screening findings are narrowing rather than aneurysm, and that a positive screen is common enough to justify scanning and rare enough that each centre needs a plan for what happens next.

Reading the angiogram

Five kinds of finding appear on these scans. The most common is hypoplasia, agenesis or stenosis of the internal carotid artery. It is often compensated by collateral vessels, but it can progress, so it calls for a repeat scan, a review by a neurologist and education of the family about the warning signs of stroke. The second is dolichoectasia of the basilar or internal carotid artery. It is a marker of a weak wall and often travels with aneurysm, so

the radiologist should search hard for an aneurysm and the blood pressure should be well controlled. The third is the saccular aneurysm, found on the basilar and vertebrobasilar arteries, the middle cerebral artery and the posterior communicating artery. Rupture can be the first and the last event, so any aneurysm goes to a neurovascular team, who decide between observation, endovascular coiling and surgical clipping. The fourth is moyamoya arteriopathy, affecting the end of the internal carotid artery and the beginning of the middle and anterior cerebral arteries. It is progressive and causes transient ischaemic attacks and strokes, so it calls for perfusion imaging, which shows how well the brain is supplied, and referral for surgical revascularisation. The fifth group is venous anomalies, such as a persistent falcine sinus or a developmental venous anomaly, which are usually incidental and need no action. One rule deserves emphasis: read the posterior circulation with care, because in adults with Alagille syndrome most of the aneurysms that ruptured were vertebrobasilar.

Do the lesions progress?

The two main series disagree, and the disagreement is informative. In Emerick's 2005 series, vascular abnormalities appeared to worsen in four of twenty-six children, and the pattern resembled moyamoya. More sobering, two asymptomatic children whose screening scans were normal later died of ischaemic events. Ten of the twenty-six had ischaemic intracranial events, and six of those ten had a cerebrovascular abnormality.
In contrast, in the 2024 Philadelphia series by Cerron-Vela and colleagues, thirteen children had repeat imaging at a mean of about fifty-five months, and the lesions were stable in all of them except one child with moyamoya. Strokes were uncommon in children under sixteen. One child had a transient ischaemic attack from an embolus, and three had small old bleeds called microhaemorrhages at diagnosis, all of them children with tetralogy of Fallot.
The honest position is that a normal angiogram lowers risk but does not close the question, and that nobody knows the right interval for repeating it. Kamath's group said exactly that in their 2023 review. The older Philadelphia series probably came from sicker and more selected patients, and the newer one is reassuring for childhood. The sensible approach is to re-image for symptoms, before surgery and at transition, rather than on a fixed clock.

Moyamoya

The best data on treating moyamoya in Alagille syndrome come from five children at North American centres, published by Lissa Baird, Edward Smith, Binita Kamath and colleagues in the Journal of Pediatrics in 2015. In the conference abstract that preceded the paper, three of the five had already had a clinical or radiographic stroke by the time

moyamoya was diagnosed, and four of the five had had transient ischaemic attacks. All five underwent indirect revascularisation, operations that lay a scalp artery or a strip of the brain's coverings onto the surface of the brain so that new vessels grow inward. These operations are called encephaloduroarteriosynangiosis and pial synangiosis. The mean age at surgery was about two and a half years, ranging from under two to four. The published series reported two serious postoperative events, one stroke and one death from a bleed in the thalamus. Function improved among the survivors, and the authors judged revascularisation to be reasonably safe in Alagille syndrome.

The practical point is that moyamoya in Alagille syndrome behaves like primary moyamoya and presents young, often after a stroke has already happened. The transplant and the neurosurgery compete for the same small child, so they need to be sequenced jointly. Under any anaesthetic, keep carbon dioxide levels, blood pressure and hydration normal, because both low carbon dioxide and low pressure reduce blood flow to a brain that depends on fragile collaterals. Decisions about antiplatelet drugs, which are often used in moyamoya, must be weighed against varices and coagulopathy, and should be made together by neurology and hepatology.

Adolescents and adults

There are no cohort studies of adults with Alagille syndrome, but a 2023 review in Hepatology International by Ayoub, Kamath and colleagues collected the reported cases. Five adults aged twenty-one to thirty, with a median age of twenty-five, had a sudden subarachnoid haemorrhage, which is bleeding into the space around the brain, usually from a ruptured aneurysm. Two of the five, forty percent, died. Three of the five bled from vertebrobasilar aneurysms, one from a posterior communicating artery aneurysm and one from an internal carotid aneurysm. Two seventeen-year-olds near the age of transition bled from posterior communicating and superior cerebellar artery aneurysms. This matters because these patients appear to rupture decades earlier than the general population, and mostly in the posterior circulation. The transition to adult care is the moment when a screening scan is most likely to be forgotten. Pregnancy adds haemodynamic load, and straining during a spontaneous vaginal delivery raises the risk of rupture. Every adult with Alagille syndrome should have had at least one angiogram of the brain, and the adult hepatologist should know the result.

Below the neck

The same arteriopathy affects the aorta and its branches. A 2025 review of kidney and vascular involvement in Alagille syndrome, published in Pediatric Nephrology, gathered the evidence. Renal artery stenosis was found on angiography or computed tomography in two of twenty-five children, eight percent, during evaluation for transplant. In a separate

series it was found on computed tomography in six of twelve children and young adults after liver transplantation, half of them, and only one of the six had high blood pressure. Across published cases, high blood pressure was present when renal artery stenosis or mid-aortic syndrome was diagnosed in seventy-nine percent, and another large-artery lesion was present alongside in seventy-eight percent, involving the coeliac, superior mesenteric, hepatic, splenic, subclavian or iliac arteries. Coarctation of the aorta was reported in four of two hundred patients in the Philadelphia cardiac cohort and two of forty-three in another series, and mid-aortic syndrome in one of those forty-three. Looking from the other direction, JAG1 variants were found in four of thirty-five families of children with mid-aortic syndrome.

So, measure blood pressure at every visit and in both arms, because a difference between the arms points to subclavian stenosis. In Alagille syndrome, hypertension is renovascular until the renal arteries have been imaged. After transplant, do not blame tacrolimus first.

The transplant vessels

The hepatic artery is part of the same disease, and this has direct consequences for liver transplantation. The 2023 review by Ayoub and colleagues recommends triphasic computed tomography of the abdomen before transplant, a scan taken in arterial, portal venous and delayed phases, to map the coeliac, hepatic, superior mesenteric and renal arteries, because these lesions are usually silent. In fifty-five children from published series, about sixty-four percent needed an aortic conduit, a graft that brings arterial blood directly from the aorta, to reconstruct the hepatic arterial supply. With a policy of conduit reconstruction, hepatic artery thrombosis fell from about thirty-five percent to about six percent. Vascular thrombosis after transplant was reported in about a quarter of children, compared with about eight percent of adults with Alagille syndrome, which fits with small vessels and weak walls.

In practice, brain angiography and abdominal computed tomography angiography belong in the same pre-transplant work-up and, in a small child, can be done under one sedation. The surgeon should hear about the arteriopathy before the operation rather than after the first abnormal Doppler scan, because a conduit planned in advance is easier than a thrombosed artery on day three. And recall that four of the eight bleeding deaths in the Paris series followed surgery.

Whom to screen, and when

No graded guideline exists, so the best we can do is make one published expert schedule explicit and explain its logic. The schedule described by the Toronto group in 2023 has five moments. At any age, a child with headache, seizure, weakness, change in speech, a transient ischaemic attack, or a head injury followed by vomiting needs emergency

imaging: magnetic resonance imaging with angiography, or computed tomography with computed tomography angiography. At diagnosis, measure blood pressure in both arms, perform a renal ultrasound, check the INR, and ask about a family history of aneurysm or stroke. At about eight years of age, perform an elective brain MRI with angiography, which at that age can usually be done without sedation. Before liver transplantation or any other major surgery, perform brain angiography together with abdominal computed tomography angiography. And at transition, soon after transfer to adult care, perform angiography if it has never been done, and again before a planned pregnancy.

The honest position is that the effectiveness of screening has never been tested, as GeneReviews says plainly. Choosing age eight trades the vascular deaths that cluster around two years against the cost and risk of sedating toddlers. It is reasonable, and this is a judgement rather than evidence, to scan earlier whenever a child is already being sedated or imaged for another reason, has hypertension, or has a family history of aneurysm or haemorrhage. The interval after a normal scan is unknown.

When the scan is abnormal

A saccular aneurysm goes to a neurovascular team, and its size, site and growth decide whether it is watched, coiled or clipped. Blood pressure control becomes strict, contact sport is avoided, and the Toronto review suggests assisted vaginal or caesarean delivery for women with intracranial vasculopathy. Moyamoya calls for perfusion imaging and referral for revascularisation before or around the time of transplant, with attention to hydration and normal carbon dioxide levels under anaesthesia, and antiplatelet drugs only by joint decision. An isolated narrowing or hypoplasia of the carotid artery calls for neurological review, repeat angiography before surgery or with symptoms, and teaching the family the signs of stroke and the importance of treating dehydration early. Renal artery stenosis or mid-aortic syndrome calls for computed tomography or catheter angiography and review by nephrology and interventional radiology, with blood pressure targets and caution with angiotensin converting enzyme inhibitors if both renal arteries are narrowed. A normal angiogram should be repeated before major surgery and at transition, and any new neurological symptom still means emergency imaging. The underlying rule is that a vascular lesion turns hepatology decisions into joint decisions about transplant timing, anaesthesia, anticoagulation and delivery.

What the guidelines say

The guidance on this subject is thin. The 2014 practice guideline from the American Association for the Study of Liver Diseases, the American Society of Transplantation and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition on evaluating children for liver transplantation asks for careful assessment of extrahepatic

disease in Alagille syndrome but does not specify brain or abdominal vascular imaging. The European Society for Paediatric Gastroenterology, Hepatology and Nutrition has no guideline on Alagille syndrome. The 2024 clinical practice guideline on genetic cholestatic liver diseases from the European Association for the Study of the Liver lists arterial disease, including disease of the brain vessels, among the features of the syndrome, but makes no recommendation about how to look for it.

Comparing the questions one by one gives a clear pattern. For screening of asymptomatic children, there is a gap: no society addresses it, the Toronto group screens at about eight, and GeneReviews notes that the benefit has never been tested. For brain imaging before transplant, the sources differ: the transplant guideline is silent on imaging, while the Toronto review screens every candidate. For mapping the abdominal arteries before transplant, they also differ: nothing is specific to Alagille syndrome in the guidelines, while the expert review recommends triphasic computed tomography and planning for a conduit. For imaging when neurological symptoms appear, the expert sources are aligned: GeneReviews says to pursue it aggressively and the Toronto review says to do it emergently. For contact sport and delivery planning there is again a gap, filled only by expert advice. And on who should treat an aneurysm or moyamoya in this syndrome, and when, there is a material gap, with nothing beyond case series. In short, no society tells hepatologists whom to scan, when or how often, even though vascular events cause about a seventh of deaths. The expert sources agree with each other; what is missing is a graded recommendation.

Common traps

Several errors recur. A child with Alagille syndrome who vomits after a head injury is sent home because the INR and platelets look close to normal. Normal coagulation tests are taken as permission for a liver biopsy that will not change management. A headache is attributed to poor sleep from itching, or to the bile acid transporter inhibitor, without imaging. A normal angiogram at the age of three is treated as a lifelong all-clear. Blood pressure is measured in one arm only, or hypertension after transplant is blamed on tacrolimus without imaging the renal arteries. Only the anterior circulation is examined closely, although adult ruptures in this syndrome have been mostly vertebrobasilar. A teenager transfers to adult care with no angiogram on record and no one owning the result. A pregnancy or a delivery is planned without knowing the vascular, portal and cardiac status. And a child is listed for transplant without mapping the hepatic artery, so the surgeon meets a hypoplastic artery in the operating theatre.

Ten things to remember

Non-cardiac vascular anomalies or events were found in nine percent of two hundred and sixty-eight patients in Philadelphia, and caused thirty-four percent of deaths in that cohort. In the GALA registry, non-cardiac vascular causes accounted for fifteen percent of one hundred and eight deaths, at a median age of two point two years. Significant bleeding occurred in twenty-two percent of one hundred and seventy-four children without liver failure, with normal median platelets and prothrombin time, and eight died, four of them after surgery. Screening angiography was abnormal in twenty-three percent of asymptomatic children in one series and fifty-six percent of an imaged cohort in another. The commonest lesion is narrowing or hypoplasia of the carotid artery, not aneurysm. Dolichoectasia travels with aneurysm, as in three of four children in the Cincinnati series. Moyamoya presents young and often after a stroke, and indirect revascularisation helped, at the cost of one stroke and one death among five children. Adults with Alagille syndrome have ruptured aneurysms at a median age of twenty-five, mostly in the posterior circulation, with forty percent of reported cases fatal. Renal artery stenosis was found in half of the children imaged after transplant, mostly without hypertension. And the working schedule is to screen with angiography at about eight, before major surgery, at transition, and at once for symptoms.

What to do differently in clinic

Return to the girl in the opening case. She needs intravenous vitamin K, a computed tomography scan of her head tonight and overnight observation, because bleeds in this syndrome occur with or without a known lesion, and coagulopathy and trauma add up. Her blood pressure should be measured in both arms and her renal arteries imaged. Since she is being imaged anyway, it makes sense to add an angiogram of her brain now rather than wait for the age of eight. She should stay away from contact sport, and her vascular status should be written into every future plan for transplant and for transition to adult care.

More broadly, a few habits follow from this chapter. Treat the arteries of the brain, the kidneys and the liver as part of the same disease as the branch pulmonary arteries. Remember that vascular deaths come early, at around two years, so the levers that do not need a scanner, vitamin K, careful procedures, blood pressure and head protection, matter from the first visit. Do not trust normal coagulation tests before a procedure. Screen with angiography at about eight, before major surgery or transplant, and at transition, and image at once for any neurological symptom. Map the hepatic artery before transplant. And make sure someone owns the result when the patient moves to adult care.

One in eleven has a lesion outside the heart, and vascular events cause about one in seven deaths. A bleed needs a weak wall plus a push. Screen at eight, before surgery, at transition, and now if there are symptoms.